

It Takes the Job — and Knows Its Limits

A cognitive task agent that senses widely, charges itself, and clears every action through a permission gate.

SIM-ONLY

— THE PROBLEM

Field autonomy is either babysat or reckless

- Supervise an autonomous robot constantly and you haven't saved any labour.
- Let it run unsupervised and it acts on stale beliefs, into hazards, or past its battery.
- And “the robot decided to” is never a defensible answer to an owner or a regulator.

The barrier isn't capability. It's whether you can hand it a job and trust it.

Capability outran governance

- **Robots can sense and plan**
rich multimodal perception is no longer the hard part.
- **But they don't self-govern**
they don't know when to ask, wait, or stop — or how to manage their own energy.
- **What's missing**
a task agent that takes direction, respects hard limits, and explains itself.

A task agent built to be governed

- **It takes a job**
an owner defines the task, the allowed and forbidden actions, and the risk ceiling.
- **It senses widely**
many modalities fuse into one world model that knows its own confidence.
- **It clears every action**
a pre-execution permission-and-harm gate rules before anything happens.

It is deliberately not an animal – no hunger, no foraging. The owner's task is the driver.

— SIX MODES

It's always in exactly one mode

MODE	WHEN
TASK_MODE	Executing the owner's task
CHARGING_MODE	Energy low — dock and preserve state
SLEEP_OPTIMIZATION_MODE	While charging — consolidate & self-test
FREE_INVESTIGATION_MODE	No task — explore safely
IDLE_LEARNING_MODE	No task — replay logs, calibrate
EMERGENCY_SAFE_MODE	Critical fault — stop safely

Every action clears the gate first



A candidate is simulated and checked against permission and harm before any verdict that lets it run.

Far more than allow or deny

- **APPROVED · VETOED · REPLACED**
run it, block it, or substitute a safe alternative.
- **ASK_OWNER**
ambiguous or risky? escalate to a human instead of guessing.
- **WAIT_FOR_INFORMATION**
too uncertain to act safely? hold and gather more.
- **LOW_ENERGY_PAUSE**
not enough charge? pause and dock before failing mid-task.

`EMERGENCY_STOP` covers the critical fault. Every verdict is logged with its reason.

The task never outranks safety

- **1 — Safety & permission**

hard harm and authorization constraints come first, always.

- **2 — Owner task**

the job runs only within those limits — never above them.

- **3 — Energy**

continuity and charging outrank optional work.

- **4 — Investigation & learning**

safe exploration and self-improvement fill the rest.

An unsafe task doesn't get a faster path. It gets a veto or a clarification request.

It senses like a field robot

18

sensor modalities fused into one world model

lidar to radiation

Per input

confidence, noise, calibration & source health

Conflicts

flagged, never averaged away

It charges itself — and picks up where it left off

OUT OF CHARGE MID-TASK?

- Predicts it will run low before finishing.
- Preserves task state, docks, enters CHARGING_MODE.

WHILE IT CHARGES

- Enters SLEEP_OPTIMIZATION_MODE.
- Consolidates memory, compresses its model, self-tests.
- Resumes the exact task on wake.

What it does in the hard cases

SCENARIO	WHAT IT DOES
unsafe_owner_task	Risky request vetoed or escalated
ambiguous_task	Asks the owner — doesn't guess
radiation_anomaly	Marks the hazard, avoids it, reports
privacy_sensitive_region	Redacts — privacy-safe summary only
low_energy_task_pause_resume	Pauses, charges, resumes, completes
sensor_conflict	Waits for information

— WHY THIS WINS

Directed, bounded, self-sufficient

APPROACH	ACTS ON ITS OWN	KNOWS WHEN TO ASK / STOP	MANAGES OWN ENERGY
Teleoperated robot	No — human drives	n/a	Manual
Scripted autopilot	Within the script	No	Limited
Unconstrained autonomy	Yes	No	No
AutonomousMind	Yes — within limits	Yes — ask / wait / pause	Yes — charges & resumes

“

An autonomous agent earns the job when it can take direction, refuse what's unsafe, ask when it's unsure, and charge itself without being told.

The case for AutonomousMind

JNEOPALLIUM • AUTONOMOUSMIND V2

rakovpublic/jneopallium

Hand it the job. Trust the limits.

A cognitive task agent with a pre-execution permission gate, energy autonomy, and a full audit trail.

```
scripts/demo-autonomous-mind/run_demo.sh
```

SIM-ONLY • 12 systems • open source

